

$$\int \frac{1}{1+x^n} dx$$

$$= \int \frac{1}{\prod_{k=1}^n (x - t_k)} dx \quad t_k = e^{i \frac{2k+1}{n} \pi}$$

where

$$\frac{1}{\prod_{k=1}^n (x - t_k)} = \sum_{k=1}^n \frac{A}{x - t_k}$$

consider $x \rightarrow t_m$

$$1 = \sum_{j=1}^n A_j \frac{\prod_{j=1}^n (x - t_j)}{x - t_k}$$

$$= A_m (t_m - t_1) \cdots (t_m - t_{m-1})(t_m - t_{m+1}) \cdots (t_m - t_n)$$

$$= A_m \frac{1+x^n}{x - t_1}$$

$$= A_m \cdot n \cdot t_1^{n-1} \quad \text{L'Hospital}$$